

BabyCare: An Automotive RF Platform for Child Presence Detection and Wireless Alerting

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Abstract—This work presents BabyCare, a compact multi-functional RF platform designed for automotive child-presence detection and thermal-risk alerting. The proposed demonstrator combines RFID-based seat verification, continuous-wave radar sensing for respiration-induced motion detection, and Bluetooth Low Energy (BLE) wireless alert transmission. The system integrates custom-designed antennas, commercially available RF modules, and an embedded processing unit into a single educational and reproducible platform suitable for classroom demonstrations and rapid prototyping. Experimental validation demonstrates the capability of the proposed system to detect child-seat occupancy and trigger real-time alerts under simulated hot-car conditions. The complete platform was developed using low-cost components while remaining within the budget constraints of the IEEE AP-S Student Design Contest.

Index Terms—Automotive antennas, Bluetooth, child-presence detection, CW radar, RFID, RF sensing.

I. INTRODUCTION

Child presence in parked vehicles represents a critical safety issue when cabin temperatures rise rapidly and no adult supervision is present. According to multiple transportation safety studies, thermal conditions inside a closed vehicle can become life-threatening within minutes, particularly for infants and young children [1]–[5].

Several approaches have been proposed for automotive child-presence detection, including pressure sensors, computer vision systems, ultrasonic sensing, and radar-based monitoring [5]. However, many existing systems either suffer from privacy concerns, limited robustness, or high implementation complexity. In addition, most solutions are difficult to reproduce in educational environments or laboratory demonstrations.

This report presents BabyCare, a compact multi-functional RF demonstrator combining three complementary RF technologies:

- RFID-based child-seat verification,
- CW radar contactless respiration sensing,
- Bluetooth wireless alert transmission.

The objective is to demonstrate how multiple automotive RF functions can be integrated into a compact embedded platform using low-cost and commercially available components. The proposed setup was designed to remain understandable for non-specialists while also illustrating key antenna and RF-engineering concepts.

The main contributions of this work are:

- the integration of multiple RF automotive functionalities into a single demonstrator,
- the design and fabrication of custom planar antennas,

- the implementation of a complete detection-to-alert chain,
- the development of a reproducible educational platform suitable for classroom demonstrations.

II. OPERATING PRINCIPLES

This section provides an accessible introduction to the three RF technologies used in BabyCare, so that the system can be understood by non-specialists.

A. RFID: Identifying Objects with Radio Waves

Radio Frequency Identification (RFID) works like a contactless barcode. A *reader* sends a radio signal that powers a passive *tag* attached to an object—here, the child seat. The tag harvests energy from the incoming wave and responds by reflecting a modified signal back to the reader, a mechanism known as *backscatter modulation*. The reader decodes this reflected signal to obtain a unique identifier.

In the UHF band around 920 MHz, RFID enables reading distances of up to several meters using low-power electronics and no battery in the tag. This makes it ideal for seat-presence verification in an automotive environment, where the system must operate reliably without user interaction.

B. CW Radar: Detecting Motion Without Contact

Continuous-wave (CW) radar relies on the *Doppler effect*: when a radio wave reflects off a moving surface, its frequency shifts proportionally to the velocity of that surface. Even very small displacements—such as the chest movement caused by breathing (typically 0.5–5 mm)—produce a measurable phase shift in the reflected signal.

The radar module transmits a continuous sinusoidal wave toward the monitored region and mixes the received echo with the transmitted signal. The resulting baseband output contains low-frequency components directly related to the target's motion. A bandpass filter isolates the respiration frequency range (approximately 0.1–0.5 Hz for a resting infant), and an embedded processor evaluates whether periodic motion is present, indicating a living occupant.

This technique is entirely contactless and preserves privacy, since no image is captured—only motion is inferred from the radio signal.

C. Bluetooth Low Energy: Wireless Alerting

Bluetooth Low Energy (BLE) operates in the 2.4 GHz ISM band and is designed for short-range communication with extremely low power consumption. Unlike classic Bluetooth, BLE uses short advertising packets that allow a device to broadcast its status periodically without establishing a persistent connection.

In BabyCare, the ESP32 microcontroller acts as a BLE peripheral that broadcasts alert notifications when a thermal-risk condition is detected. A companion smartphone application scans for these packets and displays real-time status updates. This architecture requires no pairing or user configuration, making the alert mechanism robust and straightforward to operate.

III. SYSTEM OVERVIEW

Figure 1 presents the overall architecture of the BabyCare demonstrator. The system integrates RFID sensing, CW radar monitoring, and Bluetooth Low Energy (BLE) communication into a single embedded platform centered around an ESP32 microcontroller and a custom motherboard developed for low-cost reproducibility and educational use.

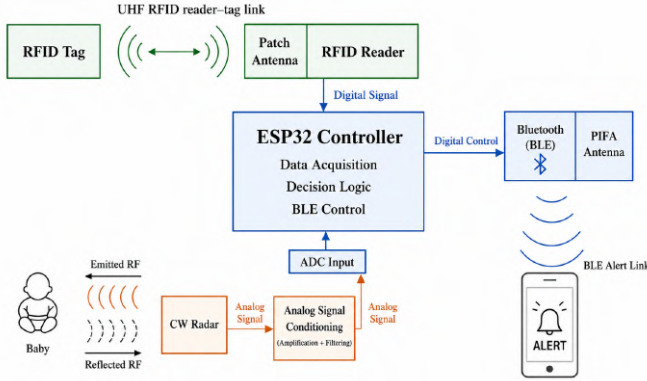


Fig. 1. Overall architecture of the BabyCare demonstrator.

The first stage of the system relies on UHF RFID technology to verify the presence of the child seat and evaluate occupancy conditions. Two RFID tags attached to the seat are interrogated through a custom 915-MHz patch antenna connected to a commercial RFID reader.

When the expected RFID conditions are satisfied, the CW radar subsystem is activated. The radar continuously illuminates the child-seat area and monitors respiration-induced motion through low-amplitude chest displacements. The received signal is processed by a custom analog conditioning board including amplification and filtering stages before digitization by the ESP32 analog-to-digital converter.

The embedded software combines information from the RFID and radar subsystems to determine whether a living occupant is present. If a potentially dangerous thermal-risk situation is detected, an alert is transmitted through a Bluetooth Low Energy (BLE) interface connected to a custom planar inverted-F antenna (PIFA). Alert notifications are then displayed on a companion smartphone application.

The overall decision process implemented by the demonstrator is summarized in Fig. 2.

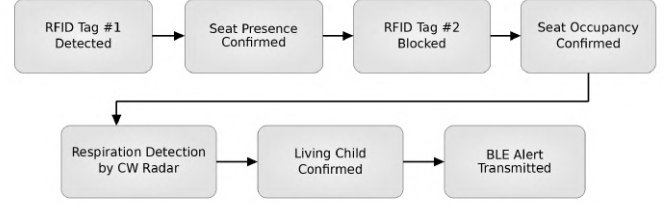


Fig. 2. Decision pipeline of the BabyCare demonstrator. RFID verifies seat presence and occupancy, CW radar confirms respiration activity, and BLE transmits warning notifications.

IV. ANTENNA SYSTEM FOR AUTOMOTIVE APPLICATIONS

A key contribution of BabyCare is the design and fabrication of custom planar antennas tailored to automotive RF applications [6]. Two antennas were developed specifically for the demonstrator: a microstrip patch antenna operating at 915 MHz for RFID-based seat verification and a planar inverted-F antenna (PIFA) operating at 2.45 GHz for Bluetooth Low Energy (BLE) communication.

Figure 3 shows the fabricated antennas integrated into the BabyCare platform. Both antennas were implemented on FR4 substrates and designed to satisfy typical automotive requirements such as low profile, compact footprint, and straightforward PCB integration.

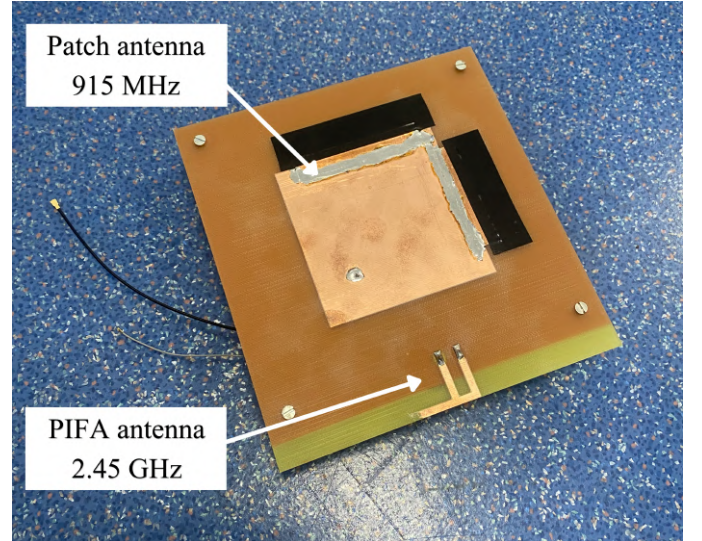


Fig. 3. Fabricated antennas used in the BabyCare demonstrator: RFID patch antenna operating at 915 MHz and PIFA operating at 2.45 GHz.

A. RFID Patch Antenna

The RFID subsystem provides the first detection stage of the BabyCare demonstrator. Two RFID tags are positioned on the child seat: a reference tag confirming seat presence and an occupancy tag used to infer the presence of an occupant

through signal blockage. The RFID reader communicates with the ESP32 via UART and continuously validates the expected seat configuration before enabling the radar subsystem.

A custom microstrip patch antenna was designed to localize the RFID interrogation region inside the vehicle mock-up. Its directional radiation pattern, low profile, and ease of PCB integration make it well suited to automotive interiors. The antenna was fabricated on a standard FR4 substrate and optimized for operation in the UHF RFID band around 915 MHz.

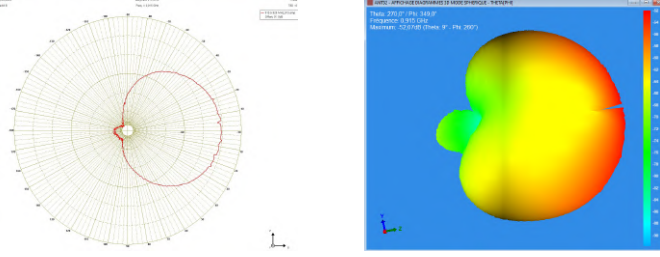


Fig. 4. Measured radiation characteristics of the RFID patch antenna at 915 MHz: (left) two-dimensional polar radiation pattern; (right) three-dimensional radiation pattern.

The measured radiation patterns confirm the directional behavior expected from a microstrip patch antenna. Both the polar cut and the three-dimensional representation exhibit a dominant forward lobe and reduced back radiation, helping to confine the RFID interrogation zone to the child-seat region. Such localization improves detection reliability while limiting unintended tag interrogation elsewhere in the vehicle. The observed radiation characteristics are therefore well suited to the intended automotive sensing application.

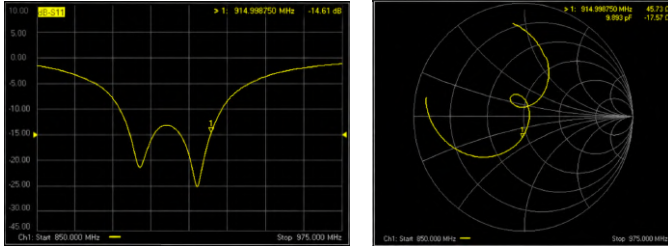


Fig. 5. Measured impedance characteristics of the RFID patch antenna: (left) reflection coefficient S_{11} (right) Smith chart representation.

Figure 5 (left) shows two resonances surrounding the target operating frequency of 915 MHz, with minima near 900 MHz and 930 MHz. Both resonances fall below -10 dB, while the measured reflection coefficient reaches -14.61 dB at 915 MHz, confirming satisfactory impedance matching for UHF RFID operation.

The Smith chart in Fig. 5 (right) provides additional insight into the antenna input impedance. At 915 MHz, the measured impedance is

$$Z = 45.7 - j17.6 \, \Omega. \quad (1)$$

The resistive component remains close to the $50 \, \Omega$ reference impedance, indicating good matching to standard RFID hardware. The small loop observed near the chart center is

consistent with the presence of the two resonances identified in the measured S_{11} response.

The directional radiation pattern provides effective coverage of the child-seat area and enables a measured RFID reading distance exceeding 2 m, well above the requirements of the proposed demonstration setup.

B. Bluetooth PIFA Antenna

The Bluetooth subsystem provides wireless communication between the BabyCare platform and the user smartphone through the integrated BLE interface of the ESP32. To support this functionality, a custom planar inverted-F antenna (PIFA) was designed for operation in the 2.4 GHz ISM band and fabricated on a standard FR4 substrate. Its compact footprint, low profile, and quasi-omnidirectional radiation characteristics make it particularly suitable for short-range in-vehicle wireless communication.

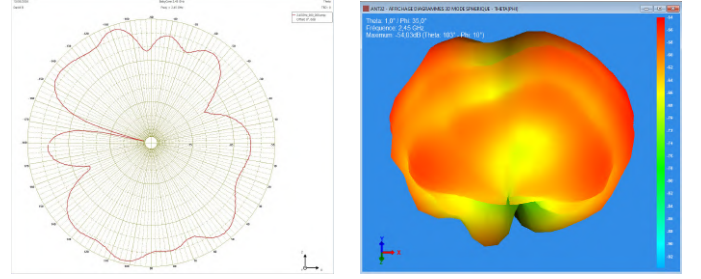


Fig. 6. Measured radiation characteristics of the BLE PIFA antenna at 2.45 GHz: (left) two-dimensional polar radiation pattern; (right) three-dimensional radiation pattern.

The measured radiation patterns confirm the expected behavior of a PIFA antenna. Both the polar cut and the three-dimensional representation exhibit a quasi-omnidirectional radiation pattern, providing reliable wireless coverage throughout the vehicle cabin regardless of the smartphone position. Such radiation characteristics are well suited to Bluetooth Low Energy communication, where broad coverage is generally preferred over highly directional operation.

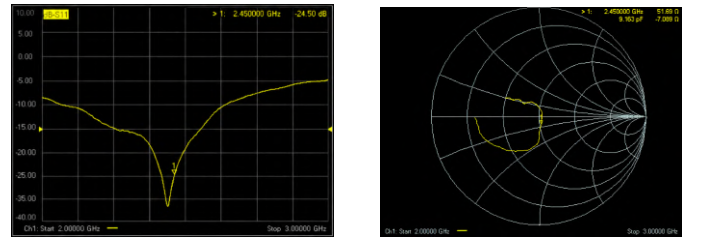


Fig. 7. Measured impedance characteristics of the BLE PIFA antenna: (left) reflection coefficient S_{11} ; (right) Smith chart representation.

Figure 7 (left) shows a well-defined resonance centered near the target frequency of 2.45 GHz. The measured reflection coefficient reaches a minimum value of -20.40 dB, indicating excellent impedance matching within the Bluetooth Low Energy operating band.

The Smith chart in Fig. 7 (right) provides additional information on the antenna input impedance. At 2.45 GHz, the measured impedance is

$$Z = 51.74 - j7.03 \, \Omega. \quad (2)$$

The resistive component remains very close to the standard $50 \, \Omega$ reference impedance, while the small reactive component indicates only minor mismatch. The smooth arc trajectory observed on the Smith chart is consistent with the single resonance visible in the measured S_{11} response.

The measured radiation characteristics and impedance matching enable reliable BLE communication throughout the demonstration setup. Experimental testing showed a communication range of approximately 100–130 m, which meets the requirements of the intended automotive application.

V. CW RADAR SENSING SUBSYSTEM

The CW radar subsystem is responsible for contactless occupant verification within the child-seat area. It serves as a second sensing stage following RFID detection and provides additional confirmation that the seat is occupied by a living passenger.

The radar module used in this work is the IPM-165 from InnoSenT, operating at 24 GHz. The module continuously illuminates the monitored region and detects motion through variations in the reflected electromagnetic signal. Even small body movements, such as respiration-induced chest motion, produce detectable changes in the received signal.

Because the radar output amplitude is relatively weak, a custom analog conditioning board was developed to amplify and filter the signal before digitization by the ESP32 analog-to-digital converter (ADC). The analog processing stage includes:

- low-noise amplification,
- bandpass filtering,
- DC offset conditioning.

The conditioned radar signal is sampled at 50 Hz and processed by a lightweight threshold-based algorithm implemented on the ESP32. The signal amplitude is continuously monitored, and variations exceeding a predefined threshold are interpreted as evidence of occupant motion within the child-seat area.

Although the radar is capable of detecting respiration-induced chest motion, the implemented decision logic does not attempt to identify the exact source of the movement. Any significant motion detected after RFID seat verification is considered sufficient evidence of occupant presence. This simplified approach was intentionally selected to improve robustness while keeping the demonstrator easy to understand and reproduce in an educational environment.

The radar subsystem therefore complements the RFID stage by distinguishing between an occupied seat and a static object placed on the seat. Experimental measurements demonstrated reliable motion detection at distances up to approximately 40 cm from the radar module, providing effective coverage of the intended child-seat monitoring region.

VI. EXPERIMENTAL VALIDATION

The complete BabyCare demonstrator was experimentally validated using a vehicle-seat mock-up reproducing realistic

child-seat conditions. The setup integrated the RFID verification stage, the CW radar sensing subsystem, the ESP32 processing unit, and the BLE communication interface into a single end-to-end platform.

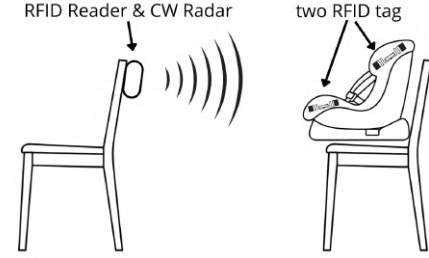


Fig. 8. Final experimental demonstration setup.

The validation procedure consisted of operating the complete detection chain under realistic demonstration conditions. The RFID subsystem first verified the presence of the child seat, after which the radar subsystem monitored the seat area for occupant motion. Detection decisions were processed by the ESP32 and warning notifications were transmitted through the BLE interface when the required conditions were satisfied.

Experimental measurements confirmed an RFID reading distance exceeding 2 m, a BLE communication range of approximately 100–130 m, and reliable radar-based motion detection within a range of approximately 40 cm. These performance levels were sufficient to demonstrate the complete detection-to-alert sequence within the intended child-seat monitoring region.

The experimental results validate the successful integration of RFID sensing, radar-based occupant verification, and Bluetooth communication into a compact automotive RF platform. The demonstrator reliably executed the complete detection pipeline and provided real-time wireless alert transmission, confirming the feasibility of the proposed approach for educational demonstrations and automotive safety applications.

VII. EDUCATIONAL VALUE AND REPRODUCIBILITY

A primary objective of BabyCare was to develop an educational platform capable of demonstrating multiple RF technologies within a single automotive application. The system combines RFID sensing, radar-based occupant verification, Bluetooth Low Energy communication, and custom antenna designs into an integrated demonstrator that remains accessible to students and non-specialists.

The project provides a hands-on introduction to several core concepts in RF and antenna engineering:

- microstrip patch antenna design and characterization,
- planar inverted-F antenna (PIFA) operation,
- UHF RFID backscatter communication,
- CW radar motion sensing,
- BLE wireless communication,
- embedded signal acquisition and decision-making.

An important educational feature of the platform is its modular architecture. Each subsystem can be operated independently as a standalone laboratory experiment. For example, the RFID stage can be used to demonstrate backscatter communication principles, while the radar subsystem illustrates contactless motion sensing and analog signal conditioning.

The demonstrator was intentionally developed using commercially available components, standard PCB fabrication techniques, and low-cost hardware. Detailed assembly instructions, software configuration procedures, and replication guidelines are provided in the companion reproduction guide submitted alongside this report.

VIII. CONCLUSION

This work presented BabyCare, a compact automotive RF platform for child-presence detection and wireless alerting. The demonstrator combines RFID sensing, CW radar occupant verification, and Bluetooth Low Energy communication using custom-designed antennas and low-cost RF hardware.

Experimental validation confirmed the feasibility of integrating multiple RF functionalities into a single platform suitable for automotive safety demonstrations and RF engineering education.

Future work will focus on improving radar robustness, antenna miniaturization, and occupant-detection capabilities.

TABLE I
BILL OF MATERIALS OF THE BABY-CARE DEMONSTRATOR.

Item	Qty	Cost (\$)	Supplier
IPM-165 Radar Module	1	6.95	Digikey
ESP32-S3-DevKitC-1U-N8R8	1	15.74	Mouser
M7E-HECTO RFID Reader	1	425.09	Mouser
RFID Tags RF600570	2	0.68	Digikey
Custom PCBs	2	100.00	PCB fabrication
3D-Printed Enclosure	1	40.00	Local fabrication
Breathing Teddy Bear	1	32.12	Amazon
Miscellaneous hardware	–	30.00	Local suppliers
Total		650.58	

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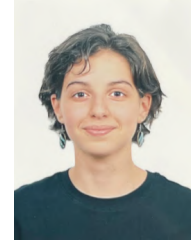
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